Mathematics

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Overview Programs Courses

CODE	NAME	DESCRIPTION
MATH100A	Mathematics for Elementary Teaching, I	This class teaches ways to think about and explain elementary school mathematics. Topics include: cultural and base-n number systems, algorithms, elementary number theory, probability, and graphing.
MATH100B	Mathematics for Elementary Teaching, II	Completes the explanation of elementary school mathematics by discussing geometry and algebra. Discusses the pedagogy with the California mathematics framework, the NCTM standards, and "replacement units".
MATH101A	Classical Number Systems	A development of classical number systems starting with natural numbers and progressing through the integral, rational, real and complex number systems. Includes an introduction to number theory, mathematical induction, the fundamental theorem of ari...
MATH101B	Mathematical Systems	The theory of operations within rings and fields. Polynomials, the fundamental theorem of algebra and other factorization theorems. Analyzing linear, polynomial, rational and exponential functions. Solving algebraic equations and inequalities. Especi...
MATH102A	Modern Euclidean and Noneuclidean	Topics in plane and solid geometry. The axioms of pure, euclidean, projective, and noneuclidean

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	Geometry	geometry. Transformational geometry (isometries, dilitations, involutions, perspectivities, and projectivities). The history and the historical implicatio...
MATH102B	Modern Euclidean and Noneuclidean Geometry	Topics in plane and solid geometry. The axioms of pure, euclidean, projective, and noneuclidean geometry. Transformational geometry (isometries, dilitations, involutions, perspectivities, and projectivities). The history and the historical implicatio...
MATH103	Introduction to Group Theory	Permutation groups, cyclic groups, theory of finite groups, group homomorphisms and isomorphisms, and Abelian groups. Applications to number theory and geometry.
MATH104A	Introduction Into Numerical Analysis	Numerical methods for the solution of nonlinear equations (Newton method), for integration (quadrature formulas and composite integration), and for the initial value problem for ordinary differential equations (Euler and Kutta methods).
MATH104B	Numerical Analysis	Numerical methods for the solution of systems of linear equations (direct and iteractive methods), and the finite difference methods for boundary value problems for (ordinary and partial) differential equations.
MATH104C	Advanced Topics in Numerical Analysis	Topics in approximation theory; numerical methods for finding eigenvalues of a matrix; and advanced topics in numerical methods for ordinary and partial differential equations.
MATH108A	Linear Algebra	Abstract vector spaces and subspaces. Span and linear independence. Basis and dimension. Linear maps. Eigenvalues and eigenvectors.
MATH108B	Linear Algebra	Diagonalization, inner product spaces, projections, least-squares approximations, invariant factors and elementary divisors, canonical forms, topics from advanced matrix theory, applied linear algebra, and group representation theory.

CODE	NAME	DESCRIPTION
MATH108C	Matrix Analysis	Eigenvalues, eigenvectors and similarity; Unitary equivalence and normal matrices; QR factorization; Least squares; Singular Value decomposition; Moore-Penrose generalized inverse; Hermitian and symmetric matrices; Variational characterization of eig...
MATH111A	Abstract Algebra	An introduction to group theory, focusing on finite groups. Basic algebraic concepts including homomorphisms, normal subgroups, quotients, direct products and group actions. Classification of finitely generated Abelian groups. Other topics such as se...
MATH111B	Abstract Algebra	Ring theory, ideals, factorization. Examples include number rings and polynomial rings. Module theory if time permits.
MATH111C	Abstract Algebra	Fields, field extensions, Galois theory.
MATH113	Non-euclidean Geometry	An introduction to hyperbolic geometry with some discussion of other non-euclidean systems.
MATH114	Computational Tools for Numerical Analysis	Introduction to programming with specific examples in scientific computing. The algorithms used will be analyzed in detail in Math 104A-B-C. The emphasis will be the translation of numerical algorithms into actual working code.
MATH115A	Introduction to Number Theory	Number-theoretic functions, divisibility, congruences, primitive roots and indices, quadratic residues and the quadratic reciprocity law.
MATH115B	Introduction to Number Theory	Selected topics at the discretion of the instructor. Possibilities include (but are not limited to) Diophantine equations, the distribution of primes, number-theoretic methods in cryptography, quadratic forms, continued fractions, algebraic number t...
MATH115C	Topics in Number Theory	Selected topics in number theory at the direction of the instructor.
MATH116	Combinatorial Analysis	Elementary counting principles, binomial coefficients, generating functions, recurrence

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		relations, the principle of inclusion and exclusion, distributions and partitions, systems of distinct representatives, applications to computation.
MATH117	Methods of Analysis	Introduction to methods of proof in analysis. topics include limits, sequences and series, continuity, compactness, as well as other topics. This course is intended to follow Mathematics 8 and to introduce students to the level of sophistication of u...
MATH118A	Real Analysis	Axioms of the real numbers, topology of metric spaces, numerical sequences, infinite series.
MATH118B	Real Analysis	Real functions: continuity, differentiability, Riemann integrals, power and trigonometric series.
MATH118C	Real Analysis	Multivariable calculus and a selection of other topics including differential forms or Lebesgue integral.
MATH119A	Ordinary Differential Equations	Existence, uniqueness, and stability; the geometry of phase space; linear systems and hyperbolicity; maps and diffeomorphisms.
MATH119B	Chaotic Dynamics and Bifurcation Theory	Hyperbolic structure and chaos; center manifolds; bifurcation theory; and the Feigenbaum and Ruelle-Takens cascades to strange attractors.
MATH122A	Theory of Complex Variables	Complex numbers, complex differentiation, analytic functions, Cauchy-Riemann equations, Taylor and Laurent series, classification of singular points, residues, evaluation of improper integrals.
MATH122B	Theory of Complex Variables	The argument principle, Rouche's theorem, analytic continuation, conformal mapping, fractional transformations and applications, infinite products, harmonic functions. Other topics at the discretion of the instructor.
MATH124A	Partial Differential Equations	Wave, heat, and potential equations.
MATH124B	Fourier Series and Numerical Methods	Fourier series; generalized functions; and numerical methods.
MATH132A	Optimization	Global and local optimization, constrained and unconstrained optimization, convex versus

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		nonconvex optimization, Projection Theorem, linear search methods, gradient methods, Newton and Quasi-Newton methods, trust region methods.
MATH132B	Operations Research	Large scale optimization, stochastic gradient descent, inexact Newton methods, limited memory BSGF, least squares problems, global optimization, optimization in general normed spaces.
MATH137A	Graph and Network Theory	Elements of graph and network theory including paths, circuits, trees, coloring, planarity, matching theory, Hall's theorem, applications to scheduling theory, flows in networks, Menger's theorem, and other topics as time permits.
MATH137B	Spectral Graph Theory	The graph Laplacian, eigenvalues of weighted graphs and random walks, the normalized Laplacian and Cheegers inequality, diameters and eigenvalues, expander graphs, eigenvalues of symmetric graphs, heat kernels.
MATH145	Introduction to Topology	Metric spaces, continuity, compactness, classification of surfaces, Euler characteristics, and fundamental groups. Additional topics at the discretion of the instructor.
MATH147A	Differential Geometry of Curves and Surfaces	Curves and surfaces in three-dimensional Euclidean space, including Frenet formulas, signed curvature of plane curves, first and second fundamental forms, normal curvature and principal curvature, Gaussian and mean curvature, Gauss equation and Theor...
MATH147B	Differential Geometry of Curves and Surfaces	Parallel transport, geodesics, Gauss-Bonnet Theorem, fundamental theorem of surfaces. Additional topics at the discretion of the instructor may include minimal surfaces, Codazzi equation, surfaces of constant Gaussian curvature, isothermal coordinate...
MATH170	Introduction to Mathematical Finance	Describes mathematical methods for estimating and evaluating asset pricing models, equilibrium and derivative pricing, options, bonds, and the term-structure of interest rates. Also introduces finance optimization models for risk management and finan...

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MATH181A	Advanced Problem Solving in Mathematical, Historical and Pedagogical Contexts.	Focuses on the representations, strategies, and language learners use to conceptualize and develop fundamental ideas of mathematics. Includes advanced mathematical problem solving and its implications for teaching and learning at the secondary level....
MATH181B	Advanced Problem Solving in Mathematical, Historical, and Pedagogical Contexts	Continuation of Math 181A or ED 134. Focuses on the representations, strategies, and language learners use to conceptualize and develop fundamental ideas of mathematics. Includes advanced mathematical problem solving and its implications for teaching...
MATH182	History of Mathematics	An examination of the major achievements in mathematical thinking throughout history. Topics may include the history of numerical systems in early civilizations, the development of formal proof, mathematical contributions from diverse populations and...
MATH190	Special Topics in Mathematics	An advanced treatment of mathematics content appropriate for undergraduates. The material for the course may come from algebra, geometry, analysis, applied mathematics or other areas of mathematics. Topics vary by instructor.
MATH190AI	Mathematical Reasonings of AI such as GPT and DeepSeek	This project-based course explores the mathematical reasonings of AI systems like GPT and DeepSeek. Students will work on exercises drawn from topics in Math 8, linear algebra, and calculus. For each project, the student will obtain an AI-generated s...
MATH190PS	Problem Solving	The Problem Solving in Mathematics course is designed to impart students with advanced problem-solving techniques and strategies in mathematics. Students will explore a diverse range of methods, including the application of parity, coloring, ordering...
MATH191	Mathematics Transition for Transfer Students	Provides transfer students with a solid mathematical foundation to ease their transition into a four-year college environment. Students will explore the fundamental axioms of integers and their properties,

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		gaining a deep understanding of this essenti...
MATH193	Internship in Mathematics	Faculty sponsored academic internship in industrial or research firms.
MATH194GS	Group Studies for Advanced Students	Participants will select a math-related book or papers, read the section before the next meeting and discuss reading at the meeting. Readings may include biographies of mathematicians, histories or popularizations of mathematics, textbooks, and readi...
MATH197A	Senior Thesis	Independent research under the supervision of a faculty member which will result in a senior thesis. Student will concentrate on reading and gathering material for a thesis.
MATH197B	Senior Thesis	Independent research under the supervision of a faculty member which will result in a senior thesis. Student will concentrate on writing a thesis.
MATH199	Independent Studies in Mathematics	Coursework consists of academic research supervised by a faculty member on a topic not available in established course offerings.
MATH199RA	Independent Research Assistance	Coursework consists of faculty supervised research assistance.
MATH201A	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH201B	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH201C	Real Analysis	Measure theory and integration. Point set topology. Principles of functional analysis. Lp-spaces. The Riesz representation theorem. Topics in real and functional analysis.
MATH202A	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex

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		analysis.
MATH202B	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex analysis.
MATH202C	Complex Analysis	Analytic functions. Complex integration. Cauchy's theorem. Series and product developments. Entire functions. Conformal mappings. Topics in complex analysis.
MATH206A	Matrix Analysis and Computation	Graduate level-matrix theory with introduction to matrix computations. SVD's, pseudoinverses, variational characterization of eigenvalues, perturbation theory, direct and iterative methods for matrix computations.
MATH206B	Numerical Simulation	Linear multistep methods and Runge-Kutta methods for ordinary differential equations: stability, order and convergence. Stiffness. Differential algebraic equations. Numerical solution of boundary value problems.
MATH206C	Numerical Solution of Partial Differential Equations--Finite Difference Methods	Finite difference methods for hyperbolic, parabolic and elliptic PDE's, with application to problems in science and engineering. Convergence, consistency, order and stability of finite difference methods. Dissipation and dispersion. Finite volume met...
MATH206D	Numerical Solution of Partial Differential Equations - Finite Element Methods	Weighted residual and finite element methods for the solution of hyperbolic, parabolic and elliptic partial differential equations, with application to problems in science and engineering. Error estimates. Standard and discontinuous Galerkin methods.
MATH214A	Ordinary Differential Equations	Existence, uniqueness, and stability; the geometry of phase space; linear systems and hyperbolicity; maps and diffeomorphisms.
MATH214B	Chaotic Dynamics and Bifurcation Theory	Hyberbolic structure and chaos; center manifolds; bifurcation theory; and the Feigenbaum and Ruelle-Takens cascades to strange attractors.

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MATH215A	Partial Differential Equations	Wave, heat, and potential equations.
MATH215B	Fourier Series and Numerical Methods	Fourier series; generalized functions; and numerical methods.
MATH220A	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH220B	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH220C	Modern Algebra	Group theory, ring and module theory, field theory, Galois theory, other topics.
MATH221A	Foundations of Topology	Metric spaces, topological spaces, continuity, Hausdorff condition, compactness, connectedness, product spaces, quotient spaces. Other topics as time allows.
MATH221B	Homotopy Theory	Homotopy groups, exact sequences, fiber spaces, covering spaces, van Kampen Theorem.
MATH221C	Differential Topology	Topological manifolds, differentiable manifolds, transversality, tangent bundles, Borsuk-Ulam theorem, orientation and intersection number, Lefschetz fixed point theorem, vector fields.
MATH225A	Topics in Number Theory	Selected topics in number theory.
MATH225B	Topics in Number Theory	Selected topics in number theory.
MATH225C	Topics in Number Theory	Selected topics in number theory.
MATH227A	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.

CODE	NAME	DESCRIPTION
MATH227B	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.
MATH227C	Advanced Topics in Geometric and Algebraic Topology	Topics, varying from year to year, include piecewise linear and differential topology, manifolds, fiber bundles and fiber spaces, homotopy theory, and spectral sequences.
MATH228A	Functional Analysis	Topics in functional analysis such as operators on Hilbert space, convex analysis, fixed point theorems, distribution theory, unbounded operators.
MATH228B	Functional Analysis	Topics in functional analysis such as operators on Hilbert space, convex analysis, fixed point theorems, distribution theory, unbounded operators.
MATH231A	Lie Groups and Lie Algebras	Differentiable manifolds, definition and examples of lie groups, lie group-lie algebra correspondence, nilpotent and solvable lie algebras, classification of semi-simple lie algebras over the complexes, representations of lie groups and lie algebras,...
MATH231B	Lie Groups and Lie Algebras	Differentiable manifolds, definition and examples of lie groups, lie group-lie algebra correspondence, nilpotent and solvable lie algebras, classification of semi-simple lie algebras over the complexes, representations of lie groups and lie algebras,...
MATH232A	Algebraic Topology	Singular homology and cohomology, exact sequences, Hurewicz theorem, Poincare duality.
MATH232B	Algebraic Topology	Singular homology and cohomology, exact sequences, Hurewicz theorem, Poincare duality.
MATH236A	Homological Algebra	Algebraic construction of homology and cohomology theories, aimed at applications to topology, geometry, groups and rings. Special emphasis on hom and tensor functors; projective, injective and flat modules; exact sequences; chain complexes; derived...
MATH236B	Homological Algebra	Algebraic construction of homology and cohomology theories, aimed at applications to

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		topology, geometry, groups and rings. Special emphasis on hom and tensor functors; projective, injective and flat modules; exact sequences; chain complexes; derived...
MATH237A	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials, Riemann-Roch theorem. Special topics may include:...
MATH237B	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials, Riemann-Roch theorem. Special topics may include:...
MATH237C	Algebraic Geometry	Affine/projective varieties, Hilbert's Nullstellensatz, morphisms of varieties, rational maps, dimension, singular/nonsingular points, blowing up of varieties, tangent spaces, divisors, differentials Riemann-Roch theorem. Special topics may include:...
MATH240A	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH240B	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH240C	Introduction to Differential Geometry and Riemannian Geometry	Topics include geometry of surfaces, manifolds, differential forms, Lie groups, Riemannian manifolds, Levi-Civita connection and curvature, curvature and topology, Hodge theory. Additional topics such as bundles and characteristic classes, spin struc...
MATH241A	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of

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		gauge theories, geometric PDE, Morse theory and Flo...
MATH241B	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of gauge theories, geometric PDE, Morse theory and Flo...
MATH241C	Topics in Differential Geometry	Various topics are covered including sectional curvature and Ricci curvature, minimal submanifolds, Atiyah-Singer index theorem and eta invariant, Einstein manifold, symplectic geometry, geometry of gauge theories, geometric PDE, Morse theory and Flo...
MATH243A	Ordinary Differential Equations	Existence and stability of solutions, Floquet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling casc...
MATH243B	Ordinary Differential Equations	Existance and stability of solutions, Floq uet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling cas...
MATH243C	Ordinary Differential Equations	Existance and stability of solutions, Floquet theory, Poincare-Bendixson theorem, invariant manifolds, existence and stability of periodic solutions, Bifurcation theory and normal forms, hyperbolic structure and chaos, Feigenbaum period-doubling casc...
MATH246A	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory an Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, elliptic regularity theory.

CODE	NAME	DESCRIPTION
MATH246B	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory and Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, elliptic regularity theory.
MATH246C	Partial Differential Equations	First-order nonlinear equations; the Cauchy problem, elements of distribution theory and Sobolev spaces; the heat, wave, and Laplace equations; additional topics such as quasilinear symmetric hyperbolic systems, ellipitic regularity theory.
MATH260	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260A	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260AA	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260B	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260BB	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260C	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260CC	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260D	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260DD	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260E	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260EE	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260ES	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260F	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260G	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260GG	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260H	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional alalysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260HH	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260I	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260II	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260J	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260JJ	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260K	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260KK	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260L	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260LL	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260M	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260MM	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260N	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260NN	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260O	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260OO	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260P	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Q	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260QQ	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260R	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260S	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260SS	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260T	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260TT	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260U	Foundations in Mathematics	Topics in algebra, analysis, applied mathematics combinatorial mathematics,functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260UU	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260V	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.

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MATH260W	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260X	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Y	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH260Z	Seminars in Mathematics	Topics in algebra, analysis, applied mathematics, combinatorial mathematics, functional analysis, geometry, statistics, topology, by means of lectures and informal conferences with members of staff.
MATH2A	Calculus with Algebra and Trigonometry	Math 3A with precalculus: A function approach integrating algebra, trigonometry, and differential calculus. Topics include: one-on-one and onto functions; inverse functions; properties and graphs of polynomial, rational, exponential, and logarithmic...
MATH2B	Calculus with Algebra and Trigonometry	Math 3B with precalculus: A continued review of relevant topics from precalculus (see Math 2A); integral calculus, including definite and indefinite integrals; techniques of integration, introduction to sequences and series; with applications in math...
MATH34A	Calculus for Social and Life Sciences	Introduction to differential and integral calculus with applications to modeling in the biological sciences.
MATH34B	Calculus for Social and Life Sciences	Continued study of differential and integral calculus with differential andintegral calculus with applications. Introduction to mathematical modeling with differential equations. Calculus of several variables including an introduction to partial deri...
MATH34LA	Calculus for the Life Sciences	Applications of differential and integral calculus to the life sciences. Includes mathematical foundations, biology examples, and computer programming to

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		simulate dynamic systems.
MATH34LB	Calculus for the Life Sciences	Applications of differential and integral calculus to the life sciences. Includes mathematical foundations, biology examples, and computer programming to simulate dynamic systems of more than one equation.
MATH3A	Calculus with Applications, First Course	Differential Calculus including analytic geometry, functions and limits, derivatives, techniques and applications of differentiation; introduction to integration; logarithmic and trigonometric functions.
MATH3B	Calculus with Applications, Second Course	Integral calculus including definite and indefinite integrals, techniques of integration; introduction to sequences and series; with applications in mathematics and physics.
MATH4A	Linear Algebra with Applications	Systems of linear equations, matrix algebra, determinants, vector spaces and subspaces, basis and dimension, linear transformations, eigenvalues and eigenvectors, diagonalization, and orthogonality.
MATH4AI	Inquiry Based Linear Algebra	Honors version of Mathematics 4A. Mathematical inquiry course is developed through problem solving and discovery.
MATH4B	Differential Equations	First and second order differential equations, separation of variables, linear differential equations, systems of first order equations, nonlinear differential equations and stability.
MATH4BI	Inquiry Based Differential Equations	Honors version of Mathematics 4B. Mathematical inquiry course is developed through problem solving and discovery.
MATH500	Teaching Assistant Practicum	Supervised teaching of undergraduate mathematics courses.
MATH501	Teaching Assistant Training	Consideration of ideas about the process of learning mathematics and discussion of approaches to teaching.

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MATH502	Teaching Associate Practicum	Supervised teaching of undergraduate courses.
MATH510	Reading for Area Examinations	Reading for area examinations.
MATH596	Directed Reading and Research	Directed reading and research.
MATH596AA	Directed Reading and Research	Directed reading and research.
MATH598	Master's Thesis Research and Preparation	Master's thesis research and preparation.
MATH598AA	Master's Thesis Research and Preparation	Master's thesis research and preparation.
MATH599	Dissertation Preparation	Dissertation preparation.
MATH6A	Vector Calculus with Applications, First Course	Calculus of functions of several variables, vector valued functions of one variable, derivative and integrals of vector functions, double and triple integrals, properties and applications of integrals, change of variables.
MATH6B	Vector calculus with Applications, Second Course	Scalar and vector fields, integration along paths, integration over surfaces and solid regions, integral theorems of vector calculus, Fourier series, partial differential equations.
MATH7H	Honors Seminar-Calculus	Emphasizing fundamental concepts and applications. Intended for highly motivated and well prepared students.
MATH8	Transition to Higher Mathematics	Introduction to the elements of propositional logic, techniques of mathematical proof, and fundamental mathematical structures, including sets, functions, relations, and other topics as time permits. Mastery of this material is essential for students...

CODE	NAME	DESCRIPTION
MATH94	Group Studies in Mathematics	Lectures and discussions on special topics. Designed for transfer students only.

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Computer Science

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Overview Programs Courses

CODE	NAME	DESCRIPTION
CMPSC1	Seminar on the Field of Computer Science	Overviews the potential of, and opportunities available from, the field of computer science. Topics include an overview of how computers work and the interesting ways in which computers can be applied to solve important and high-impact technological,...
CMPSC100	Introduction to Teaching Methods in Computer Science	Designed to train outstanding undergraduates for learning assistant positions in CS courses. Lecture/discussion surveys current research and best practices in CS pedagogy including student development theories, different pedagogical techniques, and m...
CMPSC11	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different programming language.
CMPSC110	Introduction to Research in Computer Science	Defining a CS research problem, finding and reading technical papers, oral communication, technical writing, and independent learning. Course participants work in teams as they apprentice with a CS research group to propose an original research probl...

CODE	NAME	DESCRIPTION
CMPSC111	Introduction to Computational Science	Introduction to the numerical algorithms that form the foundations of data science, machine learning, and computational science and engineering. Matrix computation, linear equation systems, eigenvalue and singular value decompositions, numerical opti...
CMPSC11AD	Programming Language Laboratory: ADA	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11AP	Programming Language Laboratory: APL	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11C	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11CO	Programming Language Laboratory: COBOL	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11FO	Programming Language Laboratory: Fortran	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11JA	Programming Language Laboratory	A self-paced course to allow a student who already possesses a working knowledge of at least one programming language an opportuity to learn otherlanguages of interest. Each section studies a different language. Sections are: 11PA-Pascal, 11FO-Fortra...

CODE	NAME	DESCRIPTION
CMPSC11LI	Programming Language Laboratory: LISP	A self-paced course to allow a studen who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11PA	Programming Language Laboratory: Pascal	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11PL	Programming Language Laboratory: PL/1	A self-paced course to allow a studen who already possesses a working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC11SN	Programming Language Laboratory: SNOBOL	A self-paced course to allow a student who already possesses working knowledge of at least one programming language an opportunity to learn other languages of interest. Each section studies a different language.
CMPSC130A	Data Structures and Algorithms	Data structures and applications with proofs of correctness and analysis. Hash Tables, Priority Queues (heaps); Balanced Search Trees. Graph Traversal techniques and their applications.
CMPSC130B	Data Structures and Algorithms II	Design and analysis of computer algorithms. Correctness proofs and solution of recurrence relations. Design techniques; divide and conquer, greedy strategies, dynamic programming. Applications of techniques to problems from several disciplines. NP -...
CMPSC132	Algorithms Engineering	Applying theoretical knowledge about algorithms, data structures, and complexity to performing the complete process of analyzing a computational problem, estimating the running time of suggested solutions, and implementing an effective and correct pr...

CODE	NAME	DESCRIPTION
CMPSC134	Randomized Algorithms	Covers advanced algorithmic ideas, focusing on randomized algorithms. Explores how randomization can be used to make algorithms simpler and more efficient, and to understand the probabilistic analysis of algorithms.
CMPSC138	Automata and Formal Languages	Formal languages; finite automata and regular expressions; properties of regular languages; pushdown automata and context-free grammars; properties of context-free languages; introduction to Turing machines and computability.
CMPSC140	Parallel Scientific Computing	Fundamentals of high performance computing and parallel algorithm design for numerical computation. Topics include parallel architectures and clusters, parallel programming with message-passing libraries and threads, program parallelization methodolo...
CMPSC148	Computer Science Project	Team-based project development. Topics include software engineering and professional development practices, interface design, advanced library support; techniques for team oriented design and development, testing and test driven development, and soft...
CMPSC153A	Hardware/Software Interface	Design and coding of embedded programs. Real-time constraints and management of resources. Practical I/O interfaces and protocols, polling, interrupt maintenance, atomic code, blocking and non-blocking tasks. Static and dynamic task scheduling under...
CMPSC154	Computer Architecture	Introduction to the architecture of computer systems. Topics include: central processing units, memory systems, channels and controllers, peripheral devices, interrupt systems, software versus hardware tradeoffs.
CMPSC156	Advanced Applications Programming	Advanced application programming using a high-level, virtual-machine-based language. Topics include generic programming, exception handling, automatic memory management, and

CODE	NAME	DESCRIPTION
		application development, management, and maintenance tools, third-party libr...
CMPSC16	Problem Solving with Computers I	Fundamental building blocks for solving problems using computers. Topics include basic computer organization and programming constructs: memory CPU, binary arithmetic, variables, expressions, statements, conditionals, iteration, functions, parameters...
CMPSC160	Translation of Programming Languages	Study of the structure of compilers. Topics include: lexical analysis; syntax analysis including LL and LR parsers; type checking; run-time environments; intermediate code generation; and compiler-construction tools.
CMPSC162	Programming Languages	Concepts of programming languages: scopes, parameter passing, storage management; control flow, exception handling; encapsulation and modularization mechanisms; reusability through genericity and inheritance; type systems; programming paradigms (impe...
CMPSC165A	Artificial Intelligence	Introduction to the field of artificial intelligence, which seeks to understand and build intelligent computational systems. Topics include intelligent agents, problem solving and heuristic search, knowledge representation and reasoning, uncertainty...
CMPSC165B	Machine Learning	Covers the most important techniques of machine learning (ML) and includes discussions of: well-posed learning problems; artificial neural networks; concept learning and general to specific ordering; decision tree learning; genetic algorithms; Bayesi...
CMPSC170	Operating Systems	Basic concepts of operating systems. The notion of a process; interprocess communication and synchronization; input-output, file systems, memory management.
CMPSC171	Distributed Systems	Distributed systems architecture, distributed programming, network of computers, message passing, remote procedure calls, group

CODE	NAME	DESCRIPTION
		communication, naming and membership problems, asynchrony, logical time, consistency, fault-tolerance, and recovery.
CMPSC172	Software Engineering	Software engineering is concerned with long-term, large-scale programming projects. Software management, cost estimates, problem specification and analysis, system design techniques, system testing and performance evaluation, and system maintenance....
CMPSC174A	Fundamentals of Database Systems	Database system architectures, relational data model, relational algebra, relational calculus, SQL, QBE, query processing, integrity constraints (key constraints, referential integrity), database design, ER and object- oriented data model, functional...
CMPSC174B	Design and Implementation Techniques of Database Systems	Queries and processing, optimizer, cost models, execution plans, rewriting rules, access methods, spatial indexing; transactions, ACID properties, concurrency control, serializability, two-phase locking, timestamping, logging,checkpointing, transacti...
CMPSC174N	Introduction to Databases	ER/object-oriented data modeling, relational data model, database design, query languages (algebra, calculus, SQL, QBE), updates and views, query processing, integrity constraints (key constraints, referential integrity), functional dependence, Boyce...
CMPSC176A	Introduction to Computer Communication Networks	Basic concepts in networking, the OSI model, error detection codes, flow control, routing, medium access control, and high-speed networks.
CMPSC176B	Network Computing	Focus on networking and web technologies used in the Internet. The class covers socket programming and web-based techniques that are used to build distributed applications.
CMPSC176C	Advanced Topics in Internet Computing	General overview of wireless and mobile networking, multimedia, security, multicast, quality of service, IPv6, and web caching. During

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		the second half of the course, one or more of the above are studied in greater detail.
CMPSC177	Computer Security	Introduction to the basics of computer security and privacy. Analysis of technical difficulties of producing secure computer information systems that provide guaranteed controlled sharing. Examination and critique of current systems, methods, certifi...
CMPSC178	Introduction to Cryptography	An introduction to the basic concepts and techniques of cryptography and cryptanalysis. Topics include: The Shannon Theory, classical systems, the enigma machine, the data encryption standard, public key systems, digital signatures, file security.
CMPSC180	Computer Graphics	Overview of OpenGL graphics standard, OpenGL statemachine, other 3D graphics libraries, 3D graphicspipeline, 3D transformations and clipping, colormodel, shading model, shadow algorithms, texturing,curves and curved surfaces, graphics hardware,intera...
CMPSC181	Introduction to Computer Vision	Overview of computer vision problems and techniques for analyzing the content of images and video. Topics include image formation, edge detection, image segmentation, pattern recognition, texture analysis, optical flow, stereo vision, shape represent...
CMPSC182	Multimedia Computing	Introduction to multimedia and applications. Topics include streaming media, conferencing, webcasting, digital libraries, multimedia system architectures, standards (including JPEG and MPEG), and multimedia storage and retrieval. A key emphasis is on...
CMPSC184	Mobile Application Development	An introduction to programming mobile computing devices. Students will learn about and study the shift in software development from desktop to mobile device applications. Topics will include software engineering and design practices, advances in prog...

CODE	NAME	DESCRIPTION
CMPSC185	Human-Computer Interaction	The study of human-computer interaction enables system architects to design useful, efficient, and enjoyable computer interfaces. This course teaches the theory, design guidelines, programming practices, and evaluation procedures behind effective hum...
CMPSC186	Internet of Things (IoT) Systems	The Internet of Things (IoT) embeds ordinary physical objects in our environment with digital intelligence -- via sensing, control, communications, and computing capabilities. As such, IoT advances are likely key drivers of the next computing revolut...
CMPSC188	Introduction to Natural Language Processing	This NLP course explores AI's capability to understand human language, focusing on machine learning algorithms and large language models. It covers language modeling, sentiment analysis, tagging, and machine translation, discussing lexical, syntactic...
CMPSC189A	Senior Computer Systems Project	This course is the first in the 2-course Computer Science Capstone sequence. Student teams learn about modern software design and engineering techniques and work together to use them and their past educational experiences at UCSB to analyze, specify,...
CMPSC189B	Senior Computer Systems Project	This course is the second in the 2-course Computer Science Capstone sequence. Student teams continue their CMPSC 189A effort by evolving their specifications, design, and prototypes of advanced computer science solutions and systems. The students lea...
CMPSC190	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190A	Special Topics in Computer Science:	These variable unit courses provide for the study of topics of current interest in computer science.

CODE	NAME	DESCRIPTION
	Foundations	
CMPSC190B	Special Topics in Computer Science: Software Systems	These variable unit courses provide for the study of topics of current interest incomputer science.
CMPSC190C	Special Topics in Computer Science: Programming Languages and Software Engineering	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190D	Special Topics in Computer Science: Information Management	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190DA	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DB	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DC	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DD	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...

CODE	NAME	DESCRIPTION
CMPSC190DE	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190DF	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC190E	Special Topics in Computer Science: Architectures	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190F	Special Topics in Computer Science: Networking	These variable unit coureses provide for the study of topics of current interest in computer science.
CMPSC190G	Special Topics in Computer Science: Security	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190H	Special Topics in Computer Science: Scientific Computing	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190I	Special Topics in Computer Science: Intelligent and Interactive Systems	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190J	Special Topics in Computer Science: General	These variable unit courses provide for the study of topics of current interest in computer science.
CMPSC190K	Special Topics in Computer Science: Computer Systems Modeling and Analysis	These variable unit courses provide for the study of topics of current interest in computer science.

CODE	NAME	DESCRIPTION
CMPSC190N	Special Topics in Computer Science: General	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC192	Projects in Computer Science	Projects in computer science for advanced undergraduate students.
CMPSC193	Internship in Industry	Special projects for selected students. Offered in conjunction with selected industrial and research firms and under direct faculty supervision. Written proposal and final report required.
CMPSC196	Undergraduate Research	Research opportunities for undergraduate students. Students will be expected to giveregular oral presentations, actively participate in a weekly seminar, and prepare at least one written report on their research.
CMPSC196B	Undergraduate Research	Advanced research for undergraduate students, by petition after completing a minimum of 4 units of CMPSC 196 for a letter grade. The student will propose a specific research project and make a public presentation of final results. Evaluation and grad...
CMPSC199	Independent Studies in Computer Science	Independent studies in computer science for advanced students.
CMPSC2	Computational Thinking for Scientists	Introduction to the central concepts of computer science as they apply to a wide variety of human endeavors, including natural and physical sciences. Topics that interrelate central ideas from algorithms, performance and complexity, data structures,...
CMPSC209	Logic and Applications in Computer Science	Propositional logic, first order logic, completeness, compactness, incompleteness, undecidability; selected topics from finite model theory, theorem proving, logic programming, program verification, databases, computational complexity.

CODE	NAME	DESCRIPTION
CMPSC211A	Matrix Analysis and Computation	Graduate level-matrix theory with introduction to matrix computations. SVD's, pseudoinverses, variational characterization of eigenvalues, perturbation theory, direct and iterative methods for matrix computations.
CMPSC211B	Numerical Simulation	Linear multistep methods and Runge-Kutta methods for ordinary differential equations: stability, order and convergence. Stiffness. Differential algebraic equations. Numerical solution of boundary value problems.
CMPSC211C	Numerical Solution of Partial Differential Equations--Finite Difference Methods	Finite difference methods for hyperbolic, parabolic and elliptic PDE's, with application to problems in science and engineering. Convergence, consistency, order and stability offinite difference methods. Dissipation and dispersion. Finite volume meth...
CMPSC211D	Numerical Solution of Partial Differential Equations--Finite Element Methods	Weighted residual and finite element methods for the solution of hyperbolic, parabolic and elliptical partial differential equations, with application to problems in science and engineering. Error Estimates. Standard and discontinuous Galerkin method...
CMPSC216	Level Set Methods	Mathematical description of the level set method and design of the numerical methods used in its implementations (ENO-WENO, Godunov, Lax-Friedrich, etc.). Introduction to the Ghost Fluid Method. Applications in CFD, Materials Sciences, Computer Visio...
CMPSC219	Sparse Matrix Algorithms	Theory, implementation, and applications of sparse matrix algorithms. Direct, iterative, preconditioned, and multilevel methods for solving large, sparse systems of linear equations. Tools include graph theory, linear algebra, and computer architectu...
CMPSC220	Theory of Computation and Complexity	Topics include: models of computation; time and space complexity classes (e.g., P, NP, CO-NP, and Pspace), efficient reducibilities, complete problems lower bounds; the polynomial

CODE	NAME	DESCRIPTION
		hierarchy.
CMPSC225	Information Theory	Entropy, mutual information, and Shannon's coding theorems; lossless source coding, Huffman, Shannon-Fano-Elias, and arithmetic codes; Channel capacity; rate-distortion theory, and lossy source coding; source-channel coding; algorithmic complexity an...
CMPSC230	Approximations, NP-Completeness and Algorithms	Epsilon approximations, PTAS and FPTAS. Techniques for the design of approximation algrorithms. P, NP, NP-complete problems, polynomial transformations, Turing reductions, strong NP-completeness, NP-hardness and inapproximability results. Topics in a...
CMPSC231	Topics in Combinatorial Algorithms	Advanced topics in algorithm design, including network flows, matchings in graphs, linear and integer programming.
CMPSC235	Computational Geometry	Algorithms and lower bound techniques in computational geometry; decision tree models of computation; geometric searching; point location and range search; convex hull and maxima of a point set; proximity algorithms; geometric intersections.
CMPSC24	Problem Solving with Computers II	Intermediate building blocks for solving problems using computers. Topics include intermediate object-oriented programming, data structures, object-oriented design, algorithms for manipulating these data structures and their run-time analyses. Data s...
CMPSC240A	Applied Parallel Computing	Interdisciplinary introduction to applied parallel computing on modern supercomputers. Topics include applications-oriented architectural issues, MPI, parallel MATLAB, and parallel numerical algorithms. A course project emphasizes understanding the r...
CMPSC24A	Problem Solving with Computers II	Intermediate building blocks for solving problems using computers. Topics include data structures, object-oriented design and development, algorithms for manipulating these

CODE	NAME	DESCRIPTION
		data structures and their runtime analyses. Data structures introduced includ...
CMPSC254	Advanced Computer Architecture	Advanced instruction set architectures, pipelining, dynamic scheduling, branch prediction, superscalar issue, out-of-order execution, memory-hierarchy design, advanced cache architectures and prefetching. Several real designs are dissected and simula...
CMPSC260	Advanced Topics in Program Analysis	Theoretical aspects of translation. Topics include: data flow analysis; control flow analysis; interprocedural analysis; optimization; type systems.
CMPSC263	Runtime Systems	Topics central to modern programming languages and their implementation: garbage collection; memory system performance; characteristics and optimization of object-oriented languages; type systems and type inference; run-time compilation.
CMPSC266	Formal Specification and Verification	Introduction to existing specification and verification systems, and the underlying theory and techniques of verifying the correctness of algorithms with respect to specifications. This subject can be considered as the combination of specification an...
CMPSC267	Automated Verification	Covers automated verification algorithms and tools. Topics include: temporal logics, fixpoint characterizations of temporal properties, model checking, symbolic verification, explicit-state verification, verification using automated theorem provers,...
CMPSC270	Operating Systems	Develops an understanding of operating systems and operating systems research at both a conceptual level and from an engineering perspective. Fundamental design principles, culled primarily from the research literature, motivate an analysis of previo...
CMPSC271	Advanced Topics in Distributed Systems	Covers the fundamental problems in distributed systems and the various tools used to solve them.

CODE	NAME	DESCRIPTION
		Of primary interest is the issue of fault-tolerance. Topics include event ordering, clocks, global states, agreement, fault-tolerance, and peer-to-peer s...
CMPSC272	Software Engineering	Principles of software engineering disciplines emphasizing requirements analysis, specification design, coding, testing and correctness proofs, maintenance, and management. Students use a number of software engineering tools.
CMPSC273	Data and Knowledge Bases	The focus is on the study of relational and post-relational data models and their query languages of different styles (algebraic, calculus, and deductive): complexity, expressive power, optimization, and database design.
CMPSC274	Advanced Topics in Database Systems	Topics include: data models, semantics, data integrity, database design, serializability theory, concurrency control, recovery, distributed databases.
CMPSC276	Advanced Topics in Networking	Focuses on advanced topics in networking. Topics may include, but are not limited to: Internet analysis, routing techniques, multimedia, approaches for network performance enhancements, and communication over new technologies.
CMPSC279	Advanced Topics in Security	Security analysis of networked systems. Analysis of techniques for network scanning, spoofing, hijacking, and denial-of-service attacks. Vulnerability analysis of applications and web-based systems. Intrusion detection and prevention techniques.
CMPSC280	Computer Graphics	Special topics in computer graphics including: curves and curved surfaces; visual perception of colors and color models; shading models; shadow generation; texture mapping; solid textures; stereographics; helmet-mounteddisplay; graphics hardware/arch...
CMPSC281	Advanced Topics in Computer Vision	Topics in Computer Vision, including imaging and multi-view geometry; image and video

CODE	NAME	DESCRIPTION
		segmentation; object detection, localization, and recognition; video activity recognition; machine learning models and methods applied to computer vision.
CMPSC281B	Advanced Topics in Computer Vision	Advanced topics in computer vision: image sequence analysis, spatio-temporal filtering, camera calibration and hand-eye coordination, robot navigation, shape representation, physically-based modeling, regularization theory, multi-sensory fusion, biol...
CMPSC284	Mobile Computing	Focuses on mobile computing. Topics may include, but are not limited to: mobile network characteristics, types of mobile networks, challenges and sollutions in mobile computing, and power conservation techniques.
CMPSC285	Advanced Image Synthesis	Theory and design of rendering systems for high-end image synthesis and modeling physically-based light transport. Topics include REYES-style architectures, Monte Carlo algorithms, variance reduction methods, path tracing, photon mapping, reflection...
CMPSC291	Special Topics in Computer Science, Applications	These courses provide for the study of topics of current interest in computer science applications. Topics are coded as follows: A. Applications, General; D. Database and Information Systems; G. Security and Cryptography; H. Computational Science and...
CMPSC291A	Special Topics in Computer Science, Applications, General	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291D	Special Topics in Computer Science, Applications, Database and Information Systems	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291G	Special Topics in Computer Science, Applications, Security and Cryptography	These courses provide for the study of topics of current interest in computer science applications.

CODE	NAME	DESCRIPTION
CMPSC291H	Special Topics in Computer Science, Applications, Computational Science and Engineering	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291I	Special Topics in Computer Science, Applications, Visual Computing and Interaction	These courses provide for the study of topics of current interest in computer science applications.
CMPSC291K	Special Topics in Computer Science, Applications	These courses provide for the study of topics of current interest in computer science applications. Topics are coded as follows: A. Applications, General; D. Database and Information Systems; G. Security and Cryptography; H. Computational Science and...
CMPSC292	Special Topics in Computer Science, Foundations	These courses provide for the study of topics of current interest in computer science foundations. Topics are coded as follows: C, Programming Languages and Software Engineering; F. Foundations, General; G. Security and Cryptography; H. Computational...
CMPSC292A	Special Topics in Computer Science, Foundations, Algorithms	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292C	Special Topics in Computer Science, Foundations, Programming Languages and Software Engineering	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292F	Special Topics in Computer Science, Foundations, General	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC292G	Special Topics in Computer Science, Foundations	These courses provide for the study of topics of current interest in computer science foundations. Topics are coded as follows: C, Programming Languages and Software Engineering; F.

CODE	NAME	DESCRIPTION
		Foundations, General; G. Security and Cryptography; H. Computational...
CMPSC292H	Special Topics in Computer Science, Foundations, Computational Science and Engineering	These courses provide for the study of topics of current interest in computer science foundations.
CMPSC293	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293A	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293B	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293C	Special Topics in Computer Science, Systems, Programming Languages and Software Engineering	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293D	Special Topics in Computer Science, Systems, Database and Information Systems	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293E	Special Topics in Computer Science, Systems, Computer Architecture and	These courses provide for the study of topics of current interest in computer science systems.

CODE	NAME	DESCRIPTION
	Embedded Systems	
CMPSC293G	Special Topics in Computer Science, Systems, Security and Cryptography	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293K	Special Topics in Computer Science, Systems, Machine Learning and Data Mining	These courses provide for the study of topics of current interest in computer science systems.
CMPSC293N	Special Topics in Computer Science, Systems	These courses provide for the study of topics of current interest in computer science systems. Topics are coded as follows: B. Operating Systems and Distributed Systems; C, Programming Languages and Software Engineering; D. Database and Information S...
CMPSC293S	Special Topics in Computer Science, Systems, General	These courses provide for the study of topics of current interest in computer science systems.
CMPSC32	Object Oriented Design and Implementation	Advanced topics in object-oriented computing. Topics include encapsulation, data hiding, inheritance, polymorphism, compilation, linking and loading, memory management, and debugging; recent advances in design and development tools, practices, librar...
CMPSC4	Computer Science Boot Camp	An introduction to computational thinking, computing, data management, and problem solving using computers, for non-majors. Topics include coding basics, representing code and data using a computer, and applications of computing that are important to...
CMPSC40	Foundations of Computer Science	Introduction to the theoretical underpinnings of computer science. Topics include propositional predicate logic, set theory, functions and relations, counting, mathematical induction and recursion (generating functions).

CODE	NAME	DESCRIPTION
CMPSC501	Techniques of Computer Science Teaching	An initial 1-2 day workshop on teaching techniques followed by a weekly seminar. Course emphasizes teaching skills, practical experience, and communication skills.
CMPSC502	Teaching of Computer Science	Procedures and techniques for teaching computer science gained through actual teaching of lecture courses, leading discussion sections, and/or teaching laboratories. Meetings will be held as needed to discuss problems,methods, and procedures.
CMPSC592	Group Studies: Controls, Dynamical Systems, and Computation	A series of weekly lectures given by university staff and outside experts in the fields of control systems, dynamical systems, and computation.
CMPSC593	Computer Science Graduate Tutorial	Individual studies exploring topics in computer science with a faculty advisor.
CMPSC594	Seminar in Computer Science	A seminar course offered on an irregular basis. Provides an in-depth discussion of advanced topics of general interest and broadens the scope ofknowledge in computer science.
CMPSC595	Group Studies in Computer Science	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engi...
CMPSC595A	Group Studies in Computer Science: Foundations	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595B	Group Studies in Computer Science: Software Systems	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.

CODE	NAME	DESCRIPTION
CMPSC595C	Group Studies in Computer Science: Programming Languages and Software Engineering	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595D	Group Studies in Computer Science	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595E	Group Studies in Computer Science: Architecture	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595F	Group Studies in Computer Science: Networking	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595G	Group Studies in Computer Science: Security	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595H	Group Studies in Computer Science: Scientific Computing	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595I	Group Studies in Computer Science: Intelligent and Interactive Systems	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595J	Group Studies in Computer Science: General	Instruction in these variable unit courses may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science.
CMPSC595K	Group Studies in Computer Science: Computer Systems	Instruction in these variable unit courses may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide

CODE	NAME	DESCRIPTION
	Modeling and Analysis	critical review of research in various areas of computer science.
CMPSC595L	Group Studies in Computer Science	Instruction in these variable unit course may be carried out by lecture, or by laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science.
CMPSC595N	Group Studies in Computer Science: Faculty Research Seminar	Special seminars focusing on topics of interest to faculty and graduate students. These seminars provide critical review of research in various areas of computer science.
CMPSC595R	Group Studies in Computer Science	Instruction in these variable unit course may be carried out by lecture, orby laboratory, or by a combination of the two. These courses provide critical review of research in various areas of computer science. See gcmps290AA-ZZ for list of topics. A:...
CMPSC596	Directed Research	Research, either experimental or theoretical, may be undertaken by qualified graduate students under the direction of a faculty member.
CMPSC597	Individual Studies for Master of Science Comprehensive Examinations and Ph.D Examinations	Individual studies for M.S. comprehensive examination and Ph.D.
CMPSC598	Master's Thesis Research and Preparation	For research underlying the thesis and writing of the thesis.
CMPSC599	Ph.D. Dissertation Research and Preparation	Research and preparation of dissertation.
CMPSC5A	Introduction to Data Science 1	Introduction to data science methods and Python programming language for students with little to no experience in the subjects. Topics include foundational programming concepts, problem-solving strategies and code design, and data science concepts as...

CODE	NAME	DESCRIPTION
CMPSC5B	Introduction to Data Science 2	Students explore the data science lifecycle, including question formulation, data collection and cleaning, exploratory data analysis and visualization, statistical inference and prediction, and decision-making. The course focus is on transforming and...
CMPSC64	Computer Organization and Logic Design	Assembly language programming and advanced computer organization; Digital logic design topics including gates, combinational circuits, flip-flops, and the design and analysis of sequential circuits.
CMPSC8	Introduction to Computer Science	Introduction to computer program development for students with little to no programming experience. Basic programming concepts, variables and expressions, data and control structures, algorithms, debugging, program design, and documentation.
CMPSC9	Intermediate Python Programming	Intermediate topics in Computer Science using the Python programming language. Topics include object oriented programming, runtime analysis, data structures, and software testing methodologies.
CMPSC90	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC90DA	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC90DB	Special Topics in Computer Science	Provides for the study of topics of current interest in computer science: A. Foundations; B. Software Systems; C. Programming Languages and Software Engineering; D. Information

CODE	NAME	DESCRIPTION
		Management; E. Architecture; F. Networking; G. Security; H. Scientific Co...
CMPSC96	Undergraduate Research	Research opportunities for undergraduate students. Students will be expected to give regular oral presentations, actively participate in a weekly seminar, and prepare at least one written report on their research.
CMPSC99	Independent Studies in Computer Science	Independent studies in computer science for advanced students.
CMPSCW8	Introduction to Computer Science	Introduction to computer program development for students with little to no programming experience. Basic programming concepts, variables and expressions, data and control structures, algorithms, debugging, program design, and documentation.
CMPSCW9	Intermediate Python Programming	Intermediate topics in Computer Science using the Python programming language. Topics include object oriented programming, runtime analysis, data structures, and software testing methodologies.

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